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# Fast diffusion in a room temperature ionic liquid confined in mesoporous carbon

S. M. CHATHOTH<sup>1(a)</sup>, E. MAMONTOV<sup>1</sup>, S. DAI<sup>2</sup>, X. WANG<sup>2(b)</sup>, P. F. FULVIO<sup>2</sup> and D. J. WESOŁOWSKI<sup>2</sup>

<sup>1</sup> Neutron Scattering Science Division, Oak Ridge National Laboratory - Oak Ridge, TN 37831, USA

<sup>2</sup> Chemical Science Division, Oak Ridge National Laboratory - Oak Ridge, TN 37831, USA

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**Abstract** – We report a quasielastic neutron scattering study in the temperature range of 290 to 350 K of a room temperature ionic liquid, [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>−</sup>], in the bulk form and confined in the  $8.8 \pm 2.1$  nm diameter pores of a mesoporous carbon matrix. In both bulk and confined liquids, our measurements, which are sensitive to the dynamics of the hydrogen-bearing cations, detect two distinct relaxation processes related to the diffusion of the cations. We have found that the cations that do not become immobilized near the pore walls exhibit an enhanced rather than suppressed diffusivity compared to the cation diffusivity in bulk liquid. Our results provide first experimental observation of molecular diffusion in a room temperature ionic liquid in confinement which is faster than diffusion in the bulk liquid.

editor's choice

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Confining fluids in nano- or mesopores is often associated with unusual structural and dynamical phenomena [1–3]. The character of the confinement-induced changes strongly depends on the confining environments, such as the pore structures, dimensions, and nature of the pore walls (*e.g.* hydrophobic *vs.* hydrophilic, rough *vs.* smooth, etc.). Recently, confinement-induced changes of the properties of electrolytes have been proposed to be of high technological relevance; *e.g.*, it was observed that a huge increase in capacitance resulted from confining an electrolyte in the pores of a carbide-derived carbon (CDC) [4]. It was shown that decreasing the pore size to a value approaching the crystallographic diameter of the ion has led to a 100% increase in normalized capacitance. Dzubiella and Hansen predicted [5] that for aqueous electrolytes under hydrophobic confinement there should be a substantial decrease in the dielectric permittivity (and a concomitant increase in the mobility) in the pores smaller than the size of the ions solvation shells. The solvation shell should become highly distorted as the water is squeezed through the pore [5]. The distortions of the solvation shells in small pores of carbon

nanostructures have been reported recently [6–8]. However, actual fast motions of the confined water molecules or ions through hydrophobic channels have never been observed experimentally (although we are aware of a report of the  $\alpha$ -relaxation in pentylene glycol that is faster in nano-confinement than in the bulk by Gorbatschow *et al.* [9]). For more complex electrolytes, such as room temperature ionic liquids (RTILs), no experimental work on the diffusivity in hydrophobic confinement has been reported. A molecular dynamics (MD) simulation has suggested fast diffusion in ionic liquids confined between parallel structureless walls [10]. The anions in the ionic liquid from the three layers near to the pore walls and cations move fast through the interlayer channels parallel to the pore surface. Using quasielastic neutron scattering (QENS), here we provide the first experimental evidence of diffusion motions in a RTIL in a hydrophobic confinement environment (carbon mesopores) that are faster compared to the corresponding diffusion motions in bulk RTIL.

The RTIL of [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>−</sup>] was prepared and purified according to the method described in the literature [11,12]. A soft-templating method was used to synthesize mesoporous carbon materials [13,14]. Loading of IL into mesoporous carbon was achieved via the incipient wetness impregnation method. Typically,

<sup>(a)</sup>E-mail: mavilachaths@ornl.gov

<sup>(b)</sup>Present Address: Angstrom Materials Inc - 1240 McCook Avenue, Dayton, OH 45404, USA.

mesoporous carbon was mixed with an ethanolic solution of IL with stirring at room temperature. The amount (volume) of IL required to fill the voids of mesoporous carbon was calculated on the basis of the pore volume of the carbon matrix. After most of ethanol evaporated, the obtained paste was further dried in a vacuum oven at 80 °C to completely remove the solvent and to facilitate the filling of IL within the voids. The cation of [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>], 1-Butyl-3-Methyl-Imidazolium (bmim), bears a number of hydrogen atoms, whereas the anion, bis(trifluoromethanesulfonyl)imide (Tf<sub>2</sub>N), has no hydrogen atoms. Since hydrogen is a very strong incoherent neutron scattering element, our QENS experiment probes the self-diffusion of the cations in this RTIL. The bulk [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] was loaded in 0.1 mm thick annular cylindrical aluminum sample holder to minimize the effect of multiple scattering on the data. The mesoporous carbon selected as a confining matrix for [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] had well characterized pores of  $8.8 \pm 2.1$  nm diameter and 100 nm–1000 nm length. The thickness of the mesoporous carbon powder sample containing the confined [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] was selected so that the sample transmission matched the transmission of the bulk sample, also to minimize the effect of multiple scattering. The sample holders were sealed with indium O-rings and mounted onto a closed-cycle refrigerator. The temperature of the samples was controlled within  $\pm 0.5$  K.

The neutron scattering experiments were performed on the backscattering spectrometer (BASIS) [15] at the Spallation Neutron Source, Oak Ridge National Laboratory. The center of the incident wavelength band, the energy resolution at the elastic line and the momentum transfer range for the experiments were  $\lambda = 6.267$  Å,  $3.4$  μeV, and  $0.2$  Å<sup>-1</sup> to  $2.0$  Å<sup>-1</sup>, respectively. In this study, for the data analysis we used a dynamic range from  $-100$  μeV to  $+100$  μeV, which is free from spurious scattering. The data were collected at the baseline temperature of 4 K and five temperature set points above the melting temperature of the [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>]. The data sets collected at 4 K were used as the resolution function.

Since [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>], similar to many other RTILs, is a glass former [16], one could expect it to exhibit the structural relaxation in the energy space in the form of a Fourier-transformed stretched exponential rather than a simple Lorentzian function. Thus, our data analysis was initially performed using Fourier transformed Kohlrausch-Williams-Watts (KWW) stretched exponential function,  $I(Q, t) = f_Q \exp[-(t/\tau)^\beta]$ , where  $f_Q$  is the Debye-Waller factor,  $\tau$  is the relaxation time parameter and  $\beta$  is the stretch factor (for a simple exponential relaxation in the time space with  $\beta = 1$ , the Fourier-transformed signal in the energy space, measured by QENS, is a Lorentzian). While a single Lorentzian could not fit the data, a single KWW component with  $\beta < 1$  provided a better fit. However, two Lorentzian components yielded even better fit, which could not be further improved using either one KWW component with ( $\beta < 1$ ) and a Lorentzian, or

two KWW components with ( $\beta < 1$ ). Therefore, similar to some previous QENS studies of RTILs [17,18], we adopted expressions (eq. (1) and eq. (2)) that included a two-Lorentzian model scattering function to fit the data independently at each  $Q$  value:

$$I(E) = [x\delta(E) + (1-x)S_{QENS}(E) + (B(E) + C)] \otimes R(E). \quad (1)$$

Here  $S_{QENS}(E)$  is the model scattering function (see below),  $R(E)$  is the resolution function,  $(B(E) + C)$  is the linear background,  $\delta(E)$  is the delta function describing the elastic scattering, and  $x$  is the elastic scattering fraction. Strictly speaking, there should be no elastic scattering from bulk liquids; however, the spectra as collected contain some elastic contribution from the sample holders and the spectrometer background, and from the carbon matrix in the confined samples. Accurate subtraction of this elastic term would require precise knowledge of the sample attenuation factors and high-statistics measurements of the background and empty holder intensities. Alternatively, the amount of elastic signal that needs to be subtracted to leave only the QENS signal from the liquid could be determined empirically for each spectrum. We chose the latter approach and simply included a freely varying elastic term,  $x\delta(E)$ , in the fits, instead of subtracting the term  $xR(E)$  from the data, as often done for the purpose data beautification. Expectedly, the value of this fit parameter was small for the bulk [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>], but became much larger for the confined [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>], due to the additional elastic scattering, some from the carbon matrix, but mostly from the ions near the pore walls that became immobilized on the experiment time scale (about a nanosecond). The  $Q$ -dependence of the parameter  $x$  allows us to estimate the contribution from the ions immobilized in the pores. At the lower  $Q$  values, there is significant small-angle scattering from the carbon matrix, but, already above  $Q \sim 0.7$  Å<sup>-1</sup>, the parameter  $x$  levels off at a value of about 0.52 for the confined sample. This is a clear sign that the intensity from the carbon matrix becomes weak for  $Q > 0.7$  Å<sup>-1</sup>, above the tail of the small-angle scattering signal, as long as the  $Q$  values remain well below the first structural peak of carbon. On the other hand, for the bulk sample, the corresponding value of the parameter  $x$  levels off at about 0.08, which represents the contribution from the sample holder and the spectrometer background. Thus, for the confined sample, the relative contributions from the measurement background, carbon matrix plus immobilized ions, and still mobile ions, are 0.08, 0.44, and 0.48, respectively. Since carbon contributes little in this  $Q$  range, the ratio of 0.44 to 0.48 can be taken as the approximate ratio of the immobilized to the still mobile ions. The approximately equal number of immobilized and still mobile ions should not be misleading; if we assume that the variations in the average density across the pore diameter are not very dramatic, this equal number is achieved with the “free” (that is, still filled with

mobile RTIL) pore diameter of about 4.7 nm, out of the total pore diameter of  $8.8 \pm 2.1$  nm.

As demonstrated earlier [17,18] the two-Lorentzian model scattering function

$$S_{QENS}(E) = \left[ p \frac{1}{\pi} \frac{\Gamma_1}{E^2 + \Gamma_1^2} + (1-p) \frac{1}{\pi} \frac{\Gamma_2}{E^2 + \Gamma_2^2} \right] \quad (2)$$

can be derived assuming the presence of two diffusion processes with different relaxation times. The longer relaxation time, associated with the slower process, is related to the narrower Lorentzian with a HWHM of  $\Gamma_2$ . The shorter relaxation time is associated with the faster process and can be obtained from the difference between the HWHMs of the two Lorentzians,  $(\Gamma_1 - \Gamma_2)$  [17,18]. The parameter  $p$  is the fraction of the intensity of faster quasielastic component and its  $Q$ -dependence provides information on spatial localization. The faster and the slower processes correspond to the  $\beta$ - and  $\alpha$ -relaxation type motions, respectively, of the center of mass, both of which exhibit a  $Q$ -dependence indicative of translational motions [19,20]. Thus, even the spatially localized  $\beta$ -relaxation component can be appropriately referred to as a diffusion process. The use of the two-Lorentzian model scattering function does not necessarily mean that each of the two dynamic components is best represented by a simple Debye-type, as opposed to stretched, process. Instead, it simply means that, in the analysis of the QENS data over the particular dynamic range, it is more important to account for the presence of the two separate relaxation processes,  $\alpha$  and  $\beta$ , then for the fact that each of these two separate processes may be of stretched character.

Examples of the data and fits at a selected temperature of 290 K and a  $Q$  value of  $0.9 \text{ \AA}^{-1}$  are presented in fig. 1. It is evident that the quasielastic components fitted in the energy space are broader for the RTIL in confinement compared to those for the bulk RTIL. Thus, the confined RTIL should exhibit some shorter relaxation times compared to the bulk liquid. This is indeed the case, as evidenced by the direct comparison of the exponential Debye-like fit components normalized to unity at  $t=0$  (lower right panel in fig. 1). It should be noted that, at a first glance, the data for the confined RTIL may appear to exhibit longer relaxation times (upper right panel in fig. 1). However, this appearance is due to the presence of a sizable elastic component (a flat background) in the confined RTIL, which originates mostly from the ions immobilized near the pore walls. The ions in the middle of the pores that retain mobility in the confined state (in particular, hydrogen-bearing cations to which neutron scattering is sensitive) actually exhibit shorter relaxation times compared to the bulk state. The effect on the data appearance of the additional elastic component in the time space (a flat background) and the energy space (enhanced relative intensity of the elastic peak) is analogous, leading to the impression of longer relaxation times. However, the quantitative analysis of the fit components confirms

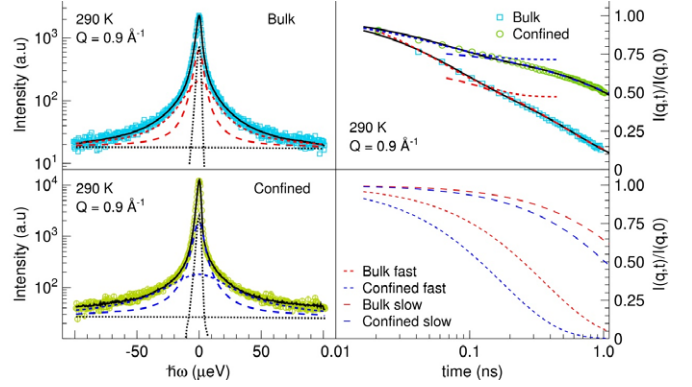


Fig. 1: (Color online) Data and fits at  $T = 290$  K and  $Q = 0.9 \text{ \AA}^{-1}$ . Left panels: the scattering intensity from [bmim]<sup>+</sup>[Tf<sub>2</sub>N]<sup>-</sup> in the bulk (upper left panel) and confined (lower left panel) states. The solid lines are the fits with a two-Lorentzian model (eq. (1) and eq. (2)). The dashed lines are the slow component, the dotted lines are the fast component and the small-dotted (black) spectrum represents the resolution function. The straight small-dotted (black) line on the bottom is the background. Right upper panel: the data and fit components (fast, slow, and overall) Fourier-transformed into the time space. Right lower panel: the fast and slow fit components normalized to unity at  $t=0$  to compare the relaxation times in the bulk and confined states.

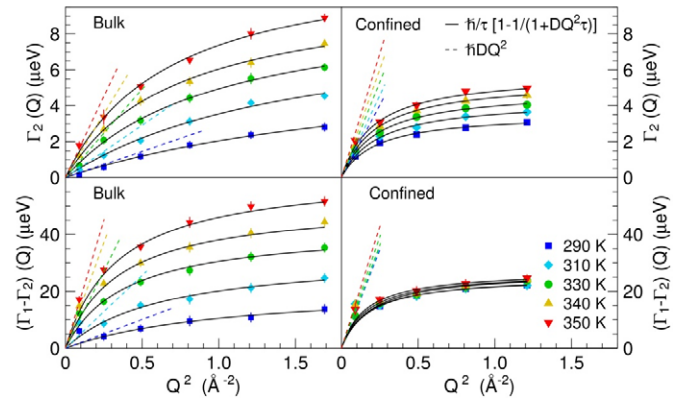


Fig. 2: (Color online) The  $Q^2$ -dependence of the half-width at half-maximum of the Lorentzian components obtained from the QENS data fits. The solid black lines are the fits of the HWHM with a jump diffusion model (eq. (3)). The straight dashed lines of the colors corresponding to the symbols colors represent the  $DQ^2$  broadening at low  $Q$  with the diffusion coefficient  $D$  obtained from the QENS data fits; the slope of these lines is proportional to the diffusion coefficient.

that those cations that remain mobile in confinement exhibit somewhat shorter relaxation times compared to their counterparts in the bulk state, from the analysis in both the energy and time space.

The observation of the diffusion processes that is faster in the confined state is corroborated by the fits of the  $Q$ -dependence of the  $\Gamma_2$  and  $(\Gamma_1 - \Gamma_2)$  (see fig. 2) with the

Table 1: The temperature dependence of the diffusion coefficients ( $D$ ), residence time ( $\tau$ ), jump length ( $\langle l \rangle$ ) of cations in [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] ionic liquid in the bulk and confined states.

| Temperature (K) | Bulk [bmim <sup>+</sup> ][Tf <sub>2</sub> N <sup>-</sup> ]         |                |                            |                                                                    |                |                            | Confined [bmim <sup>+</sup> ][Tf <sub>2</sub> N <sup>-</sup> ]     |                |                            |                                                                    |                |                            |
|-----------------|--------------------------------------------------------------------|----------------|----------------------------|--------------------------------------------------------------------|----------------|----------------------------|--------------------------------------------------------------------|----------------|----------------------------|--------------------------------------------------------------------|----------------|----------------------------|
|                 | Fast $D$<br>(10 <sup>-10</sup><br>m <sup>2</sup> s <sup>-1</sup> ) | $\tau$<br>(ps) | $\langle l \rangle$<br>(Å) | Slow $D$<br>(10 <sup>-10</sup><br>m <sup>2</sup> s <sup>-1</sup> ) | $\tau$<br>(ps) | $\langle l \rangle$<br>(Å) | Fast $D$<br>(10 <sup>-10</sup><br>m <sup>2</sup> s <sup>-1</sup> ) | $\tau$<br>(ps) | $\langle l \rangle$<br>(Å) | Slow $D$<br>(10 <sup>-10</sup><br>m <sup>2</sup> s <sup>-1</sup> ) | $\tau$<br>(ps) | $\langle l \rangle$<br>(Å) |
| 290             | 3.06 ± 0.24                                                        | 30.09          | 2.3                        | 0.44 ± 0.03                                                        | 97.10          | 1.6                        | 21.00 ± 0.46                                                       | 25.63          | 5.6                        | 2.45 ± 0.20                                                        | 183.64         | 5.1                        |
| 310             | 7.78 ± 0.86                                                        | 19.84          | 3.0                        | 0.89 ± 0.07                                                        | 73.52          | 1.9                        | 22.04 ± 0.19                                                       | 26.00          | 5.7                        | 2.78 ± 0.28                                                        | 151.62         | 5.0                        |
| 330             | 16.47 ± 1.22                                                       | 15.49          | 3.9                        | 1.50 ± 0.06                                                        | 66.76          | 2.4                        | 21.38 ± 2.07                                                       | 24.33          | 5.5                        | 3.23 ± 0.20                                                        | 134.63         | 5.1                        |
| 340             | 22.5 ± 1.26                                                        | 12.85          | 4.1                        | 2.16 ± 0.01                                                        | 62.69          | 2.8                        | 23.76 ± 3.26                                                       | 24.37          | 5.8                        | 3.66 ± 0.02                                                        | 120.67         | 5.0                        |
| 350             | 27.27 ± 1.24                                                       | 10.63          | 4.1                        | 2.76 ± 0.17                                                        | 53.21          | 2.9                        | 26.02 ± 2.38                                                       | 23.96          | 6.1                        | 4.19 ± 0.16                                                        | 113.11         | 5.3                        |

following expression for a jump diffusion model [21]:

$$\Gamma(Q) = \frac{\hbar}{\tau} \left[ 1 - \frac{1}{1 + DQ^2\tau} \right]. \quad (3)$$

Here  $\tau$  is the residence time between the diffusion jumps of a diffusing particle with a diffusion coefficient  $D = \langle l^2 \rangle / 6\tau$ , where  $\langle l^2 \rangle$  is the mean square jump length. It should be noted that the data fits with eq. (3) were limited to a  $Q$  range of 0.5–1.3 Å<sup>-1</sup>. Below  $Q = 0.5$  Å<sup>-1</sup>, QENS broadening becomes saturated due to the confined character of the fast diffusion process. Above a  $Q$  value of 1.3 Å<sup>-1</sup>, both the ( $\Gamma_1 - \Gamma_2$ ) and  $\Gamma_2$  fit parameters exhibit a pronounced narrowing of the QENS broadening. Small angle X-ray scattering results show that the first strong structure factor maximum of the [bmim<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] ionic liquid is at  $Q = 1.4$  Å<sup>-1</sup> [22]. Thus, the narrowing of the quasielastic line width above  $Q = 1.3$  Å<sup>-1</sup> must be due to the de Gennes narrowing effect [23] (the movement of a particle to the next-neighbor position in a dense fluid is hindered and needs a rearrangement of the surrounding particles, which results in a slowing-down of the relaxation process at the momentum transfer vectors of structure factor maximum. In the frequency domain one observes a reduction of the quasielastic line width is known as de Gennes narrowing) on the coherent QENS scattering that begins to contribute significantly at the position of the first strong structure factor maximum. As evidenced by the higher slope of the  $DQ^2$  straight lines shown in fig. 2 at low  $Q$  values, in confined liquid the broadening tends to increase faster at low  $Q$ , indicating higher diffusion coefficients, even though the residence times, determined from the high- $Q$  broadening, are shorter in bulk liquid. It should be noted that the  $DQ^2$  straight lines are based on the  $D$  values from the overall fits (solid lines in fig. 2), and, therefore, are not distorted by the possible effects of confinement on the QENS broadening signal at low  $Q$  values, which will be discussed below.

The values for the temperature-dependent diffusion coefficient ( $D$ ), residence time ( $\tau$ ), and mean jump length ( $\langle l \rangle$ ), are presented in table 1. We note in passing that both

narrow and broad quasielastic components obtained from our fits describe processes related to the center-of-mass dynamics, and neither of these two components is related to the much faster processes due to the dynamics of the side groups, which are present on the picosecond time scale and are too broad for the dynamic range of our experiment [18]. Likewise, we should note that both narrow and broad quasielastic components are related to the diffusion dynamics of the cations, since the non-hydrogen-bearing anions contribute little to the incoherent scattering signal. The width of the faster of the two translational processes, represented by the ( $\Gamma_1 - \Gamma_2$ ), shows some leveling off below 0.5 Å<sup>-1</sup>. This leveling off is not observed for the width of the slower translational component, represented by the  $\Gamma_2$ , which follows the  $Q^2$ -dependence at low  $Q$ . While the slower component is related to the long-range translational diffusion, the faster component describes a spatially constrained diffusion process. The confinement radius of 7–11 Å can be deduced from the low- $Q$  value of the QENS broadening of the broad component, which is expected to remain approximately constant at a value of  $4.33D/a^2$  for a translational diffusion process within a spherical volume of radius  $a$  [24]. The properties of this faster, localized diffusion process, which appears to be common in RTILs, have been discussed in detail elsewhere [17,18]. The slower process can be directly compared with the diffusion data measured by other techniques, such as NMR [25] (see fig. 3 for the temperature dependence of the diffusion coefficients and activation energies ( $E_a$ ) for the diffusion processes).

QENS is arguably the best technique to probe multiple diffusion processes due to its capability of extracting the information on the spatial characteristics of diffusion jumps through the analysis of the  $Q$ -dependence of the scattering signal. Examples of QENS studies that have revealed multiple diffusion processes in RTILs include [H<sub>2</sub>NC(dma)<sub>2</sub>][BETI] [17] [Ag(propylamine)<sub>2</sub><sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] and [Ag(1-pentene)<sup>+</sup>][Tf<sub>2</sub>N<sup>-</sup>] [18] and 1-*n*-butylpyridinium bis((trifluoromethyl)sulfonyl)imide ([BuPy][Tf<sub>2</sub>N]) [26]. Additionally, <sup>13</sup>C NMR [27], molecular



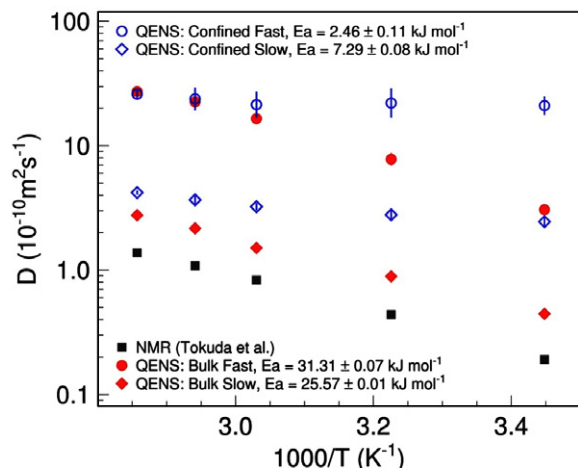


Fig. 3: (Color online) The temperature dependence of the diffusion coefficients in  $[\text{bmim}^+][\text{Tf}_2\text{N}^-]$  measured in this work using QENS. The long-range translational diffusion coefficients measured using nuclear magnetic resonance (ref. [25]) are shown for comparison.

dynamics [28,29] and dielectric spectroscopy [30] studies also have identified the existence of two dynamic processes in an ionic liquid  $[\text{bmim}^+][\text{PF}_6^-]$ . While the origin of the fast localized process remains debated, there is a possibility that it is related to the presence of clusters, or nano-scale heterogeneities [31], which are believed to be common in RTILs. For instance, neutron and X-ray diffraction studies of  $[\text{bmim}^+][\text{PF}_6^-]$  have detected a strong pre-peak associated with a short-range order [32], which was attributed to the clusters of  $\sim 11 \text{ \AA}$  in size. The typical size of these clusters is similar to the spatial localization of the fast diffusion process in the present and earlier [17,18] studies.

Importantly, both the typical cluster size and the spatial confinement of the fast diffusion process (which may actually represent one and the same phenomenon) are substantially smaller than the confinement pore size in this study ( $8.8 \pm 2.1 \text{ nm}$ ). For this reason, the confinement does not seem to qualitatively change the picture of the two simultaneous diffusion processes in  $[\text{bmim}^+][\text{Tf}_2\text{N}^-]$ . The main result of the present study is the enhancement of the translational diffusion observed under confinement, whereas the character of this diffusion remains qualitatively unchanged compared to the bulk state, in that two diffusion processes, a spatially localized and a long-range one, are still present.

Possible explanations of the enhanced diffusion compared to the bulk have been summarized in a comprehensive review by Alba-Simionesco *et al.* [33]. One reason could be the restriction on the cooperativity length imposed by the confinement [34]. It is also possible that the structural changes in the confined RTIL induced by the pore walls are responsible for the enhanced diffusion dynamics. MD simulations on similar ionic liquids on a graphite surface have indicated a dense region near the surface of the graphite, with three density oscillation

layers extending to more than  $15 \text{ \AA}$  [35,36]. Spatial heterogeneity has been found within the interfacial region, in which the polar groups aggregate to form a polar network among the imidazolium ring of the cation, while the alkyl side chain on the cation are distributed in the voids and form non-polar domains. The MD simulation has also indicated a higher relative density of the anions compared to that of the cations through the interfacial region [34]. Similar results have been obtained for an ionic liquid  $[\text{dmim}^+][\text{Cl}^-]$  confined in mesoporous carbon [10]. The lower density of the cations in the middle of the pores might be the reason for the faster motions of the cations observed in the present QENS study, which are associated with these lower density regions. In this scenario, the cations near the pore walls are immobilized, at least on the time scale of the neutron scattering experiment, and contribute only to the elastic signal. As we discussed above, estimates based on the value of the parameter  $x$  in eq. (1) suggest approximately equal numbers of the immobilized and still mobile ions for the RTIL in the confining matrix.

In conclusion, we used quasielastic neutron scattering to investigate the dynamical processes in  $[\text{bmim}^+][\text{Tf}_2\text{N}^-]$  ionic liquid, in its bulk form and confined in a mesoporous carbon with  $8.8 \pm 2.1 \text{ nm}$  cylindrical pores. In both the confined and bulk states,  $[\text{bmim}^+][\text{Tf}_2\text{N}^-]$  displays at least two dynamical processes in the time range accessible by the QENS experiment. The slower of the two processes is the long-range translational diffusion process, whereas the faster process is related to the confined diffusion spatially localized on the length scale of about  $10 \text{ \AA}$ . Remarkably, the diffusivity values that characterize both processes have been found to be comparatively faster in the confined state, at least at the lower temperatures. At higher temperatures, the difference in the diffusion coefficients between the bulk and confined states decreases or disappears. Our results provide the first evidence that confinement may enhance, rather than suppress, the diffusivity of a certain fraction of cations in RTILs.

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